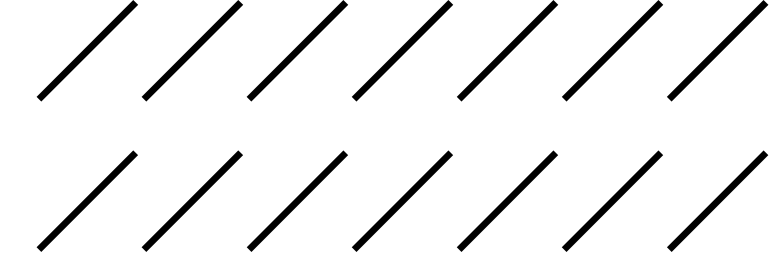


DRAFTING CLASS DIAGRAMS

AGENDA

- 
- 01 Class Diagrams - Simple Practice
 - 02 Identifying classes and attributes
 - 03 Identifying methods
 - 04 Drafting Class Diagram
 - 05 Identifying Associations
 - 06 Identifying inheritance, aggregation, and compositions
 - 07 Reviewing Class Diagram
 - 08 Linking to Next Weeks

section 1

CLASS DIAGRAMS SIMPLE PRACTICE

Drafting Class Diagrams -- Simple Practice

The SimpleClinicSys is a medical information system that intends to support a small surgery. In this system, medical doctors — or general practitioners (GPs) — are assigned to patients and they may have a substitute doctor. A doctor has a staff id, a name, an address and a contact telephone number. A patient has an NHS number, a name, an address and a contact telephone number. The system keeps a record of the visits of patients to the clinic with the following information: the doctor who has seen the patient, when the visit has taken place, a small description of what the visited consisted of. The system also keeps information on treatments that are administered to patients; a treatment includes a description of the diagnosed medical condition, the doctor that made the diagnostic and started the treatment, the date when the treatment started, the prescribed medicines, an information on whether the treatment is active, cancelled or concluded, a description on why the treatment has been closed (cancelled or concluded) and the date when it has been closed. A medicine has a code and a name.

Using a tool of your choice, draft a class diagram for SimpleClinicSys based on the brief above. A sample solution is under this week's Moodle.

section 2

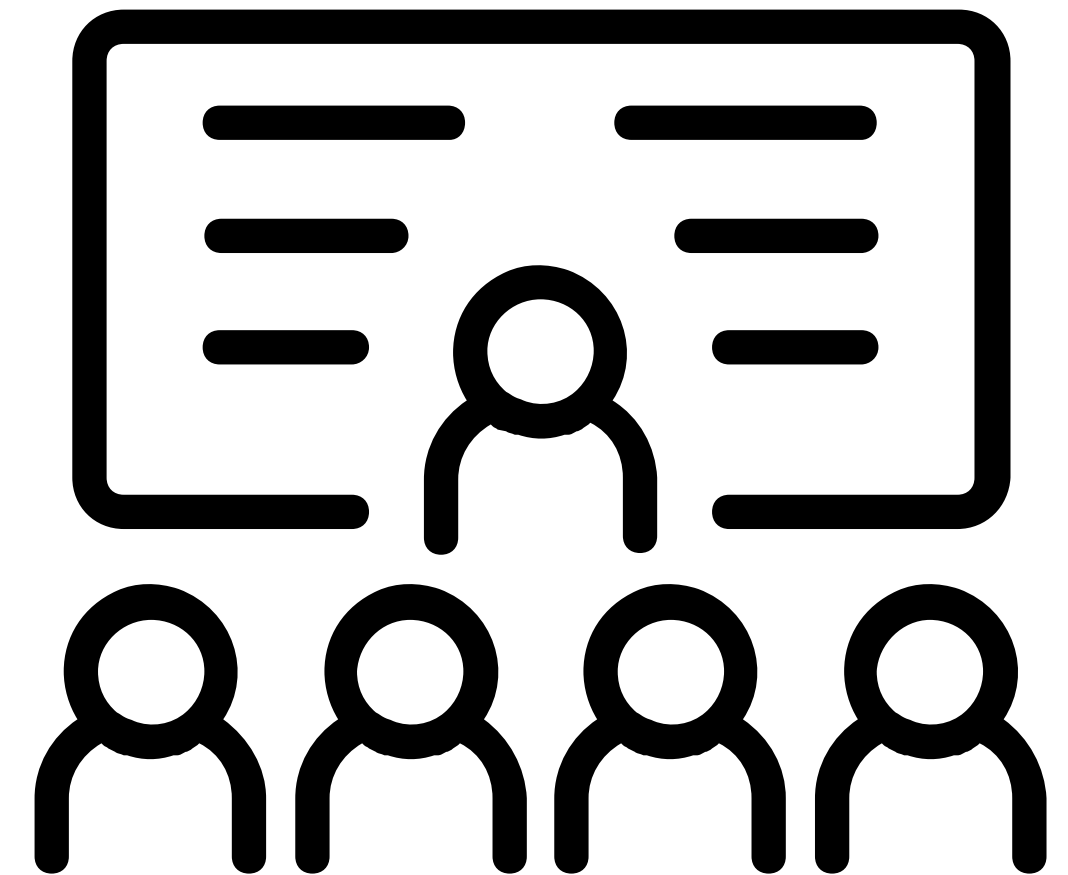
IDENTIFYING CLASSES AND ATTRIBUTES

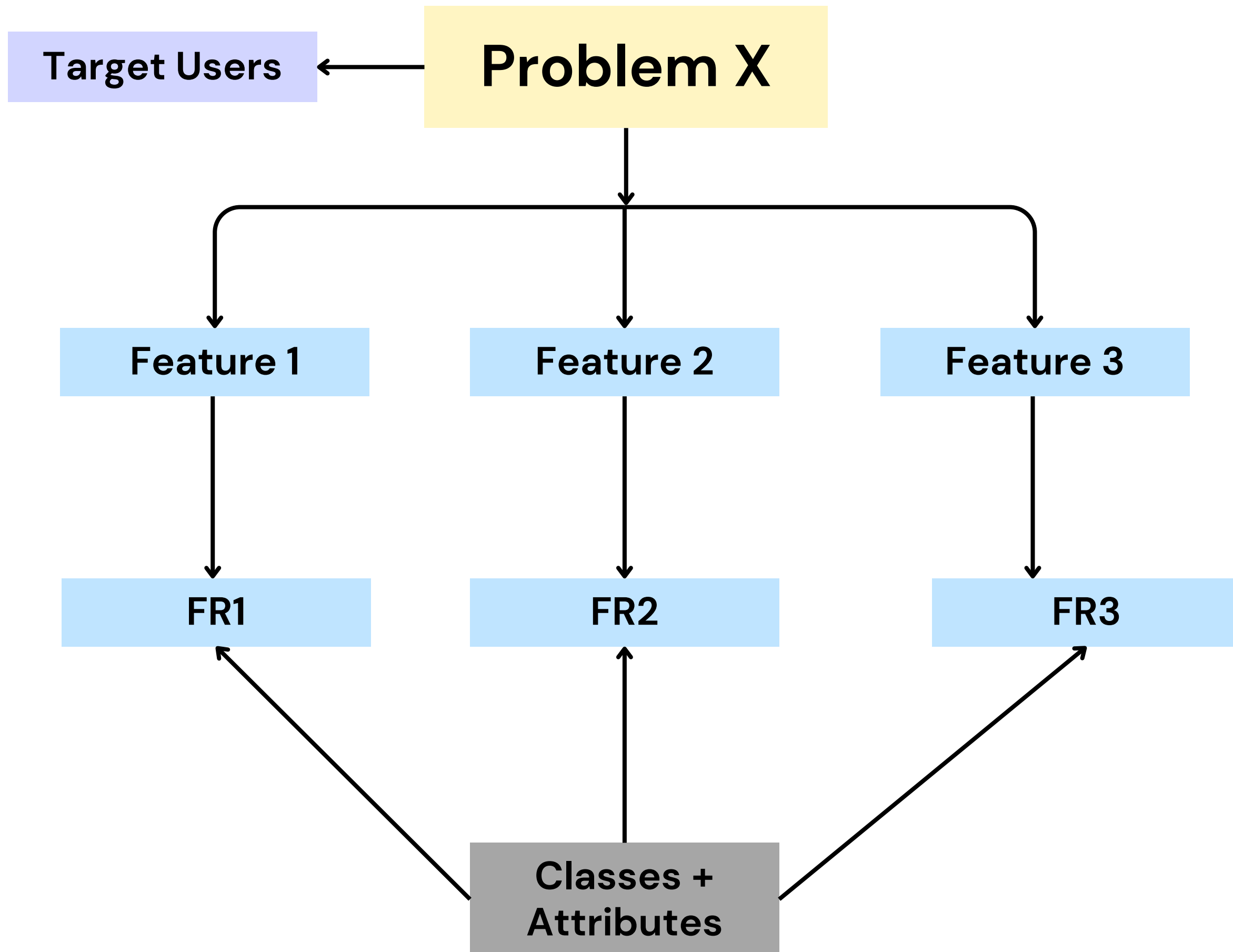
2.1. Identifying classes and attributes

The functional requirements and chosen case study brief are the first places to identify entities/classes of the system. Consistency between the class diagram and these documents is key to make the class diagram of use.

Activity

- For your chosen case study, underline all the nouns in the functional requirements table.
- For your chosen case study, categorise them as classes or attributes.
 - Can more information be held about this entity? Class.
 - Can no more information be held about it? Attribute.





section 3

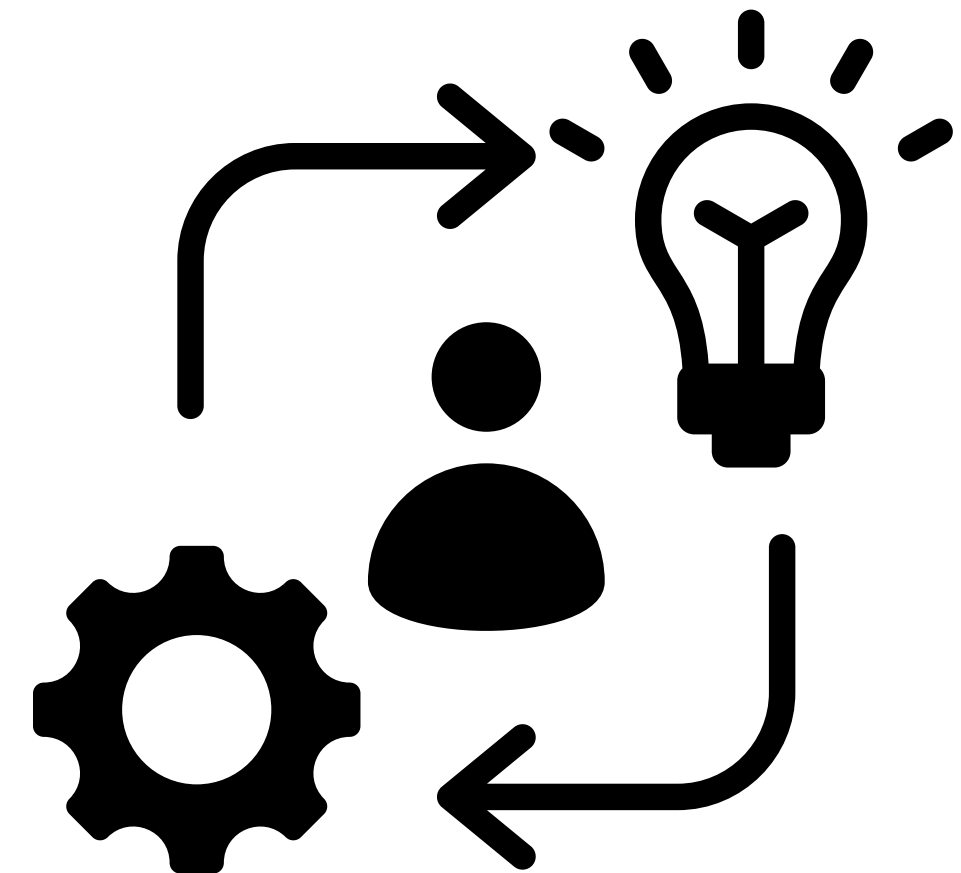
IDENTIFYING METHODS

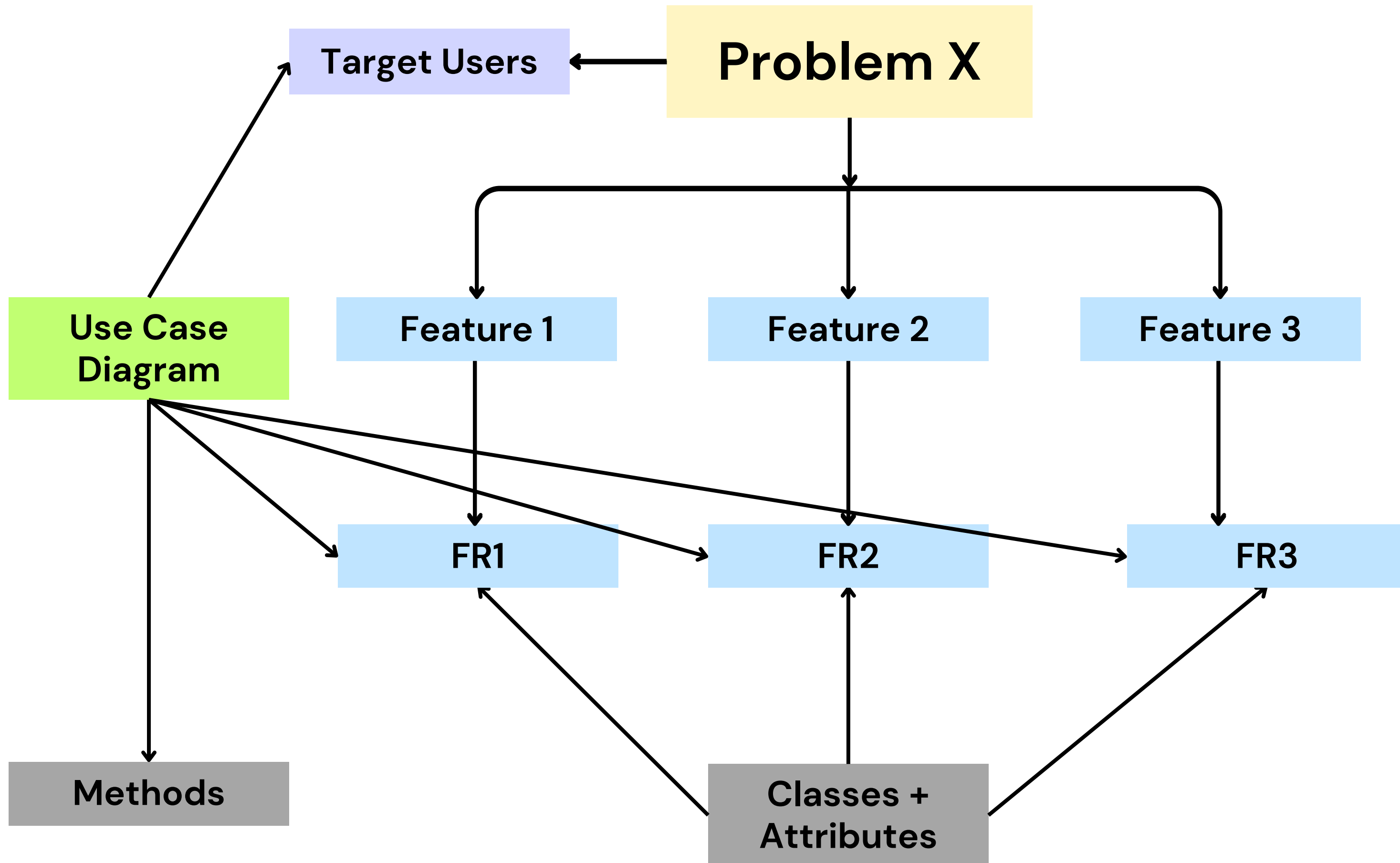
3.1. Identifying methods

The functional requirements and chosen case study brief are the first places to identify entities/classes of the system. Consistency between the class diagram and these documents is key to make the class diagram of use.

Activity

- For your chosen case study, underline all the verbs in the use case diagram you drafted.
- Categorise them in the classes where they should be implemented, not where they should be called.





section 4

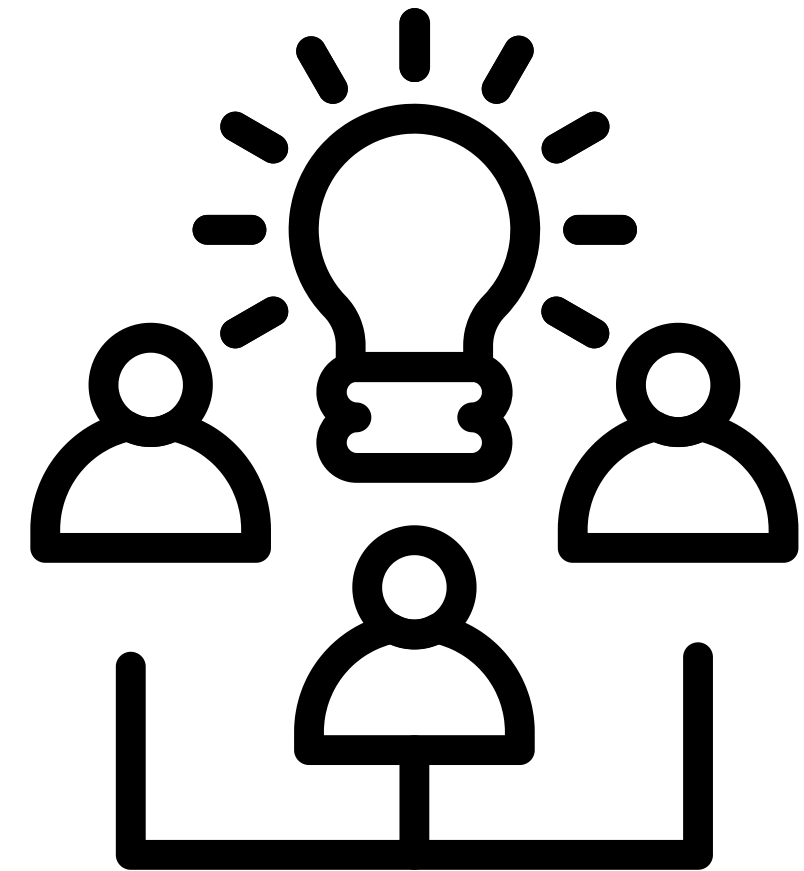
DRAFTING CLASS DIAGRAM

4.1. Drafting class diagram

Once the class, attributes, and methods are identified, they can be drawn in the class diagram for your chosen case study.

Activity

- For your chosen case study:
 - Create classes based on derived entities.
 - Add methods with their parameters and return types.
 - Add attributes that are needed to fulfil the methods.



section 5

IDENTIFYING ASSOCIATIONS

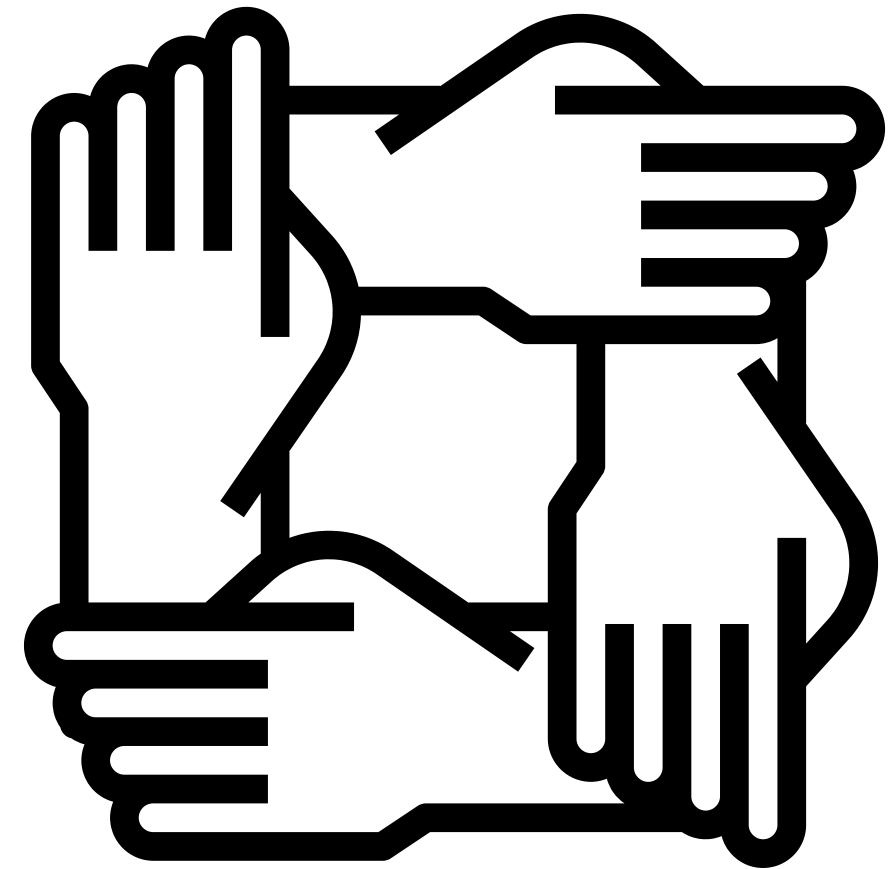
5.1. Identifying associations

The associations are needed to indicate how classes would interact with each other.

In later weeks, you will be updating this class diagram with more detailed relationships between the classes.

Activity

- For your chosen case study:
 - Use functional requirements and use case diagram arrows to identify associations.
 - Add labels on the associations.
 - Add multiplicities on the associations.
 - Add navigabilities on the associations.



section 6

**IDENTIFYING
INHERITANCE, AGGREGATION,
AND COMPOSITIONS**

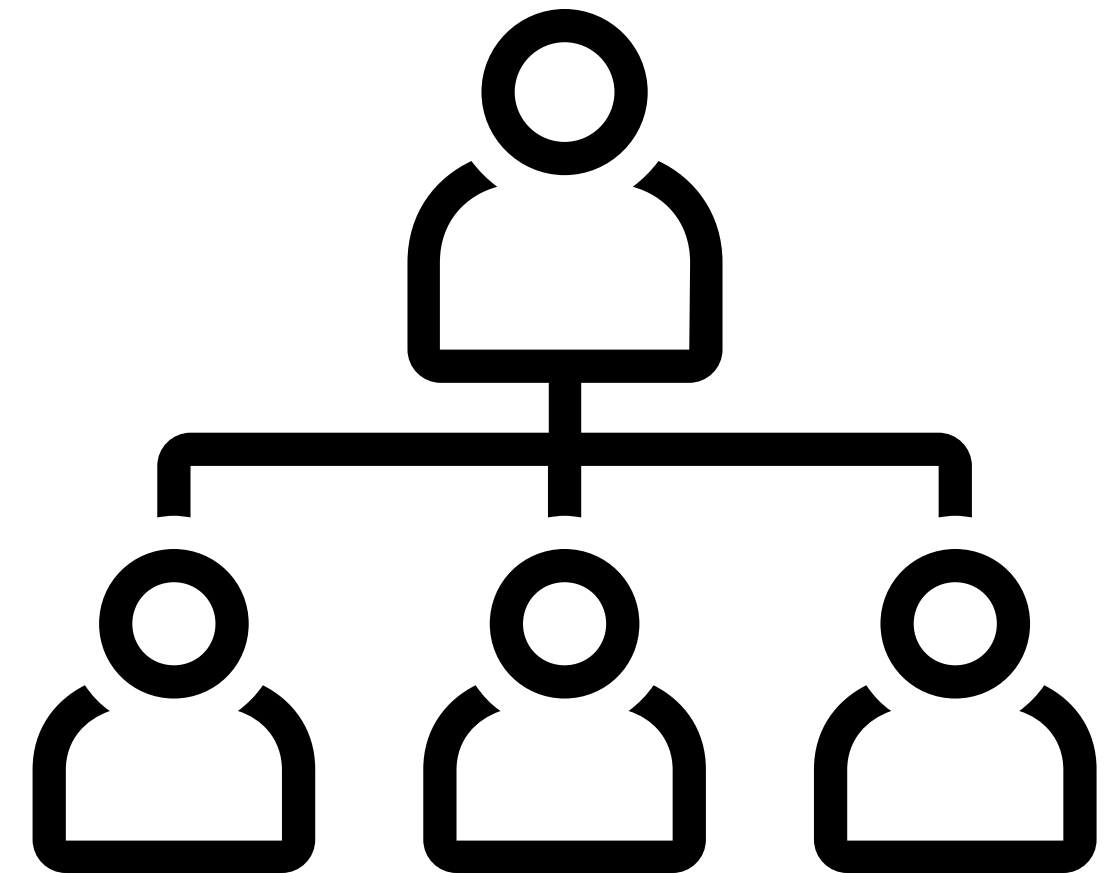
6.1. Identifying inheritance, aggregation, and compositions

Given your draft class diagram, some of the associations are better as inheritance, aggregation and/or composition.

This has implications on the implementation (e.g. making one class a sub-class of another in code).

Activity

- For each of the associations in your draft class diagram:
 - Do the associated classes have common attributes/methods? Inheritance.
 - Are they linked as whole-part but can exist without each other? Aggregation.
 - If not, composition.



section 7

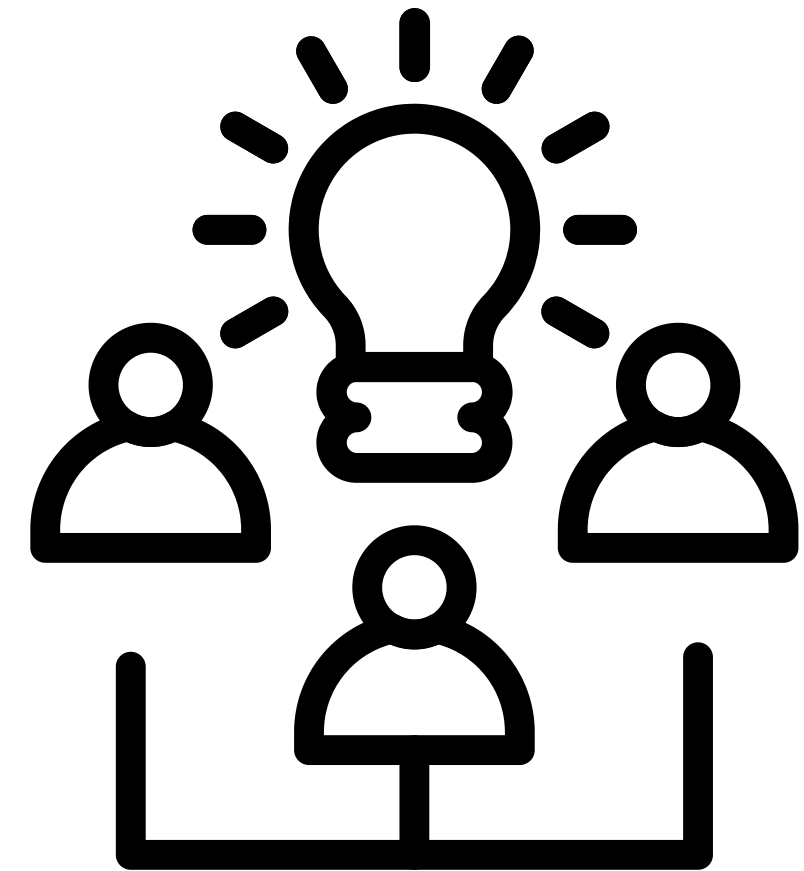
REVIEWING CLASS DIAGRAM

7.1. Reviewing class diagram

Once the changes are identified, they can be drawn in a re-drafted class diagram for your chosen case study.

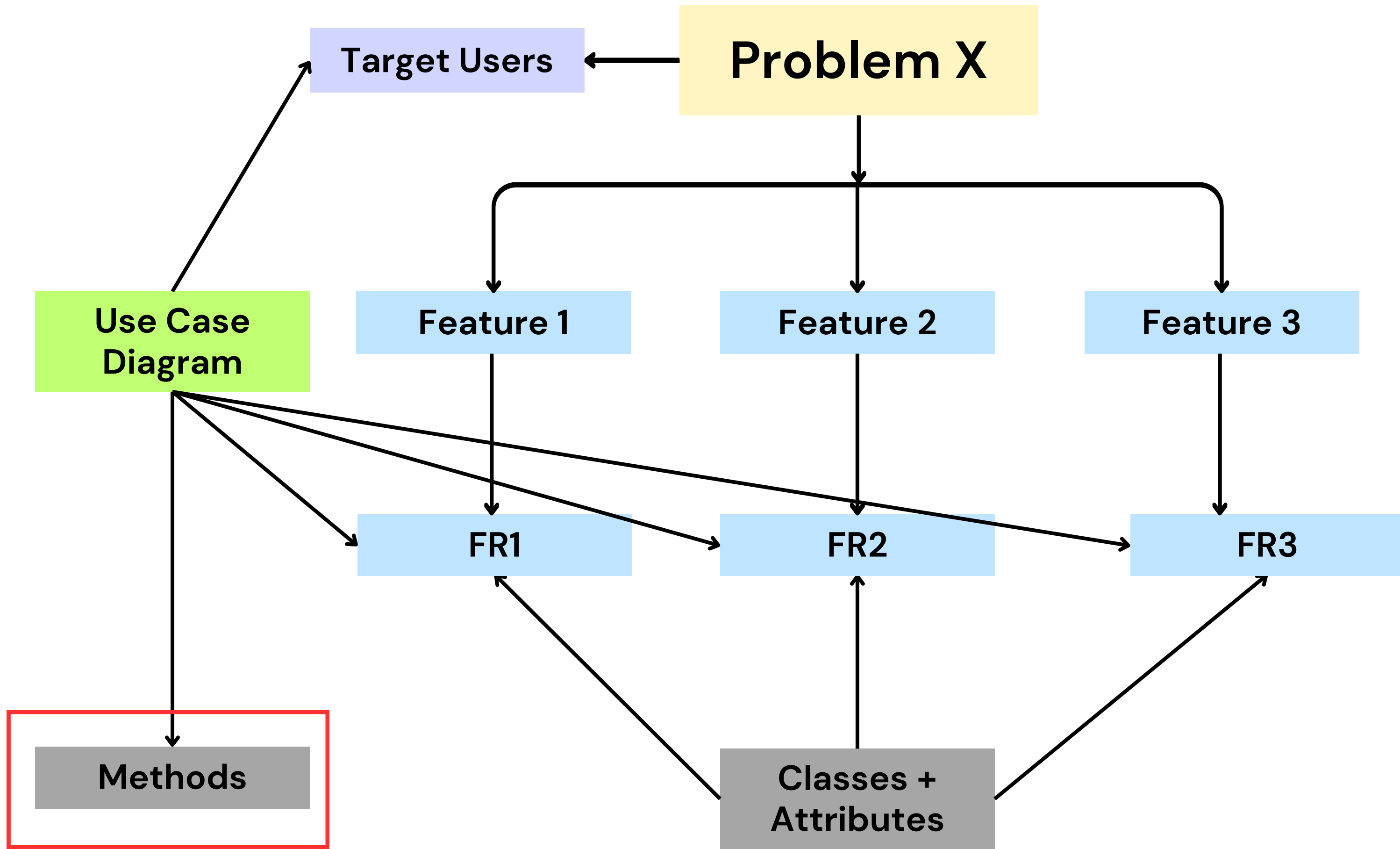
Activity

- Adjust the associations, where necessary adding classes/attributes/ and methods.
- Add any relevant multiplicities and labels.
- Review the notations of the class diagram.



section 8

LINKING TO NEXT WEEKS



Sequence Diagrams

Thank You!

ANONYMOUS FEEDBACK

